

Exercise 4.3

1. Find HCF by factorization method.

(i) $21x^2y, 35xy^2$

$$21x^2y = 3 \cdot \boxed{7} \cdot \boxed{x} \cdot x \cdot \boxed{y}$$

$$35xy^2 = 5 \cdot \boxed{7} \cdot \boxed{x} \cdot y \cdot \boxed{y}$$

$$\text{HCF} = 7xy$$

(ii) $4x^2 - 9y^2, 2x^2 - 3xy$

Factorization

$$\begin{aligned} 4x^2 - 9y^2 &= (2x)^2 - (3y)^2 \\ &= (2x + 3y)(2x - 3y) \end{aligned}$$

$$\begin{aligned} 2x^2 - 3xy &= x(2x - 3y) \end{aligned}$$

Now

$$4x^2 - 9y^2 = (2x + 3y)\boxed{(2x - 3y)}$$

$$2x^2 - 3xy = x\boxed{(2x - 3y)}$$

$$\text{HCF} = (2x - 3y)$$

(iii) $x^3 - 1, x^2 + x + 1$

Factorization

$$\begin{aligned} x^3 - 1 &= (x)^3 - (1)^3 \\ &= (x + 1)[(x)^2 + (x)(1) + (1)^2] \\ &= (x + 1)(x^2 + x + 1) \end{aligned}$$

Now

$$x^3 - 1 = (x + 1)\boxed{(x^2 + x + 1)}$$

$$x^2 + x + 1 = \boxed{x^2 + x + 1}$$

$$\text{HCF} = x^2 + x + 1$$

(iv) $a^3 + 2a^2 - 3a, 2a^3 + 5a^2 - 3a$

Factorization

$$\begin{aligned} a^3 + 2a^2 - 3a &= a[a^2 + 2a - 3] \\ &= a[a^2 + 3a - a - 3] \\ &= a[a(a + 3) - 1(a + 3)] \\ &= a[(a + 3)(a - 1)] \\ &= a(a + 3)(a - 1) \end{aligned}$$

Also

$$\begin{aligned} 2a^3 + 5a^2 - 3a &= a[2a^2 + 5a - 3] \\ &= a[2a^2 + 6a - a - 3] \\ &= a[2a(a + 3) - 1(a + 3)] \\ &= a[(a + 3)(2a - 1)] \\ &= a(a + 3)(2a - 1) \end{aligned}$$

Now

$$a^3 + 2a^2 - 3a = \boxed{a(a + 3)}(a - 1)$$

$$2a^3 + 5a^2 - 3a = \boxed{a(a + 3)}(2a - 1)$$

$$\text{HCF} = a(a + 3)$$

(v) $t^2 - 3t - 4, t^2 + 5t + 4, t^2 - 1$

Factorization

$$\begin{aligned} t^2 - 3t - 4 &= t^2 - 4t + t - 4 \\ &= t(t - 4) + 1(t - 4) \\ &= (t - 4)(t + 1) \end{aligned}$$

Also

$$\begin{aligned} t^2 + 5t + 4 &= t^2 + 4t + t + 4 \\ &= t(t + 4) + 1(t + 4) \\ &= (t + 4)(t + 1) \end{aligned}$$

Also

$$\begin{aligned} t^2 - 1 &= (t)^2 - (1)^2 \\ &= (t + 1)(t - 1) \end{aligned}$$

Now

$$t^2 - 3t - 4 = (t - 4)\boxed{(t + 1)}$$

$$t^2 + 5t + 4 = (t + 4)\boxed{(t + 1)}$$

$$t^2 - 1 = \boxed{(t + 1)}(t - 1)$$

$$\text{HCF} = t + 1$$

(v) $x^2 + 15x + 56, x^2 + 5x - 24, x^2 + 8x$

Factorization

$$\begin{aligned} x^2 + 15x + 56 &= x^2 + 8x + 7x + 56 \\ &= x(x + 8) + 7(x + 8) \\ &= (x + 8)(x + 7) \end{aligned}$$

Also

$$\begin{aligned} x^2 + 5x - 24 &= x^2 + 8x - 3x - 24 \\ &= x(x + 8) - 3(x + 8) \\ &= (x + 8)(x - 3) \end{aligned}$$

Also

$$\begin{aligned} x^2 + 8x &= x(x + 8) \end{aligned}$$

Now

$$x^2 + 15x + 56 = \boxed{(x + 8)}(x + 7)$$

$$x^2 + 5x - 24 = \boxed{(x + 8)}(x - 3)$$

$$x^2 + 8x = x\boxed{(x + 8)}$$

$$\text{HCF} = x + 8$$

2. Find HCF the following expressions by using division method:

(i) $27x^3 + 9x^2 - 3x - 10, 3x - 2$

$$\begin{array}{r}
 9x^2 + 9x + 5 \\
 3x - 2 \overline{) 27x^3 + 9x^2 - 3x - 10} \\
 \underline{\pm 27x^3 \mp 18x^2} \\
 27x^2 - 3x - 10 \\
 \underline{\pm 27x^2 \mp 18x} \\
 15x - 10 \\
 \underline{\pm 15x \mp 10} \\
 0
 \end{array}$$

$$HCF = 3x - 2$$

(ii) $x^3 - 9x^2 + 23x - 15, x^2 - 4x + 3$

$$\begin{array}{r}
 x - 5 \\
 x^2 - 4x + 3 \overline{) x^3 - 9x^2 + 21x - 9} \\
 \underline{\pm x^3 \mp 4x^2 \pm 3x} \\
 -5x^2 + 18x - 9 \\
 \underline{\mp 5x^2 \pm 20x \mp 15} \\
 -2x + 6
 \end{array}$$

$$-2x + 6 = -2(x - 3)$$

We ignore -2 because it is not common to both the given polynomials

$$\begin{array}{r}
 x - 1 \\
 x - 3 \overline{) x^2 - 4x + 3} \\
 \underline{\pm x^2 \mp 3x} \\
 -x + 3 \\
 \underline{\mp x \pm 3} \\
 0
 \end{array}$$

$$HCF = x - 3$$

(iii) $2x^3 + 2x^2 + 2x + 2, 6x^3 + 12x^2 + 6x + 12$

$$\begin{aligned}
 2x^3 + 2x^2 + 2x + 2 &= 2(x^3 + x^2 + x + 1) \\
 6x^3 + 12x^2 + 6x + 12 &= 6(x^3 + 2x^2 + x + 2) \\
 &= 2 \times 3(x^3 + 2x^2 + x + 2)
 \end{aligned}$$

$$\begin{array}{r}
 1 \\
 x^3 + x^2 + x + 1 \overline{) x^3 + 2x^2 + x + 2} \\
 \underline{\pm x^3 \pm x^2 \pm x \pm 1} \\
 x^2 + 1
 \end{array}$$

Now

$$\begin{array}{r}
 x + 1 \\
 x^2 + 1 \overline{) x^3 + x^2 + x + 1} \\
 \underline{\pm x^3 \pm x} \\
 x^2 + 1 \\
 \underline{\pm x^2 \pm 1} \\
 0
 \end{array}$$

$$HCF = 2(x^2 + 1)$$

(iv) $2x^3 - 4x^2 - 16x, x^3 - 4x, 3x^2 - 6x$

$$\begin{array}{r}
 2 \\
 x^3 - 4x \overline{) 2x^3 - 4x^2 - 16x} \\
 \underline{\pm 2x^3 \mp 8x^2} \\
 4x^2 - 16x
 \end{array}$$

$$4x^2 - 16 = 4(x^2 - 4)$$

We ignore 4 because it is not common to both the given polynomials

$$\begin{array}{r}
 x \\
 x^2 - 4 \overline{) x^3 - 4x} \\
 \underline{\pm x^3 \mp 4x} \\
 0
 \end{array}$$

Now

$$\begin{array}{r}
 3 \\
 x^2 - 4 \overline{) 3x^2 - 6x} \\
 \underline{\pm 3x^2 \mp 12} \\
 -6x + 12
 \end{array}$$

$$-6x + 12 = -6(x - 2)$$

We ignore -6 because it is not common to both the given polynomials

$$\begin{array}{r}
 x + 2 \\
 x - 2 \overline{) x^2 - 4} \\
 \underline{\pm x^2 \mp 2x} \\
 2x - 4 \\
 \underline{\pm 2x \mp 4} \\
 0
 \end{array}$$

$$HCF = x - 2$$

3. Find LCM of the following expressions by using prime factorization method:

(i) $2a^2b, 4ab^2, 6ab$

$$2a^2b = \boxed{2} \cdot a \cdot \boxed{a} \cdot \boxed{b}$$

$$4ab^2 = \boxed{2} \cdot 2 \cdot \boxed{a} \cdot \boxed{b} \cdot b$$

$$6ab = \boxed{2} \cdot 3 \cdot \boxed{a} \cdot \boxed{b}$$

$$LCM = [2 \cdot a \cdot b][2 \cdot 3 \cdot a \cdot b]$$

$$LCM = 12a^2b^2$$

(ii) $x^2 + x, x^3 + x^2$

Factorization

$$\begin{aligned} x^2 + x \\ = x(x + 1) \end{aligned}$$

$$\begin{aligned} x^3 + x^2 \\ = x^2(x + 1) \end{aligned}$$

Now

$$\begin{aligned} x^2 + x &= \boxed{x} \cdot \boxed{(x + 1)} \\ x^3 + x^2 &= \boxed{x} \cdot x \cdot \boxed{(x + 1)} \\ LCM &= [x(x + 1)][x] \\ LCM &= x^2(x + 1) \end{aligned}$$

(iii) $a^2 - 4a + 4, a^2 - 2a$

Factorization

$$\begin{aligned} a^2 - 4a + 4 \\ = a^2 - 2a - 2a + 4 \\ = a(a - 2) - 2(a - 2) \\ = (a - 2)(a - 2) \end{aligned}$$

$$\begin{aligned} a^2 - 2a \\ = a(a - 2) \end{aligned}$$

Now

$$\begin{aligned} a^2 - 4a + 4 &= \boxed{(a - 2)}(a - 2) \\ a^2 - 2a &= a\boxed{(a - 2)} \\ LCM &= [(a - 2)][a \cdot (a - 2)] \\ LCM &= a(a - 2)^2 \end{aligned}$$

(iv) $x^4 - 16, x^3 - 4x$

Factorization

$$\begin{aligned} x^4 - 16 \\ = (x^2)^2 - (4)^2 \\ = (x^2 + 4)(x^2 - 4) \\ = (x^2 + 4)[(x)^2 - (2)^2] \\ = (x^2 + 4)(x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} x^3 - 4x \\ = x(x^2 - 4) \\ = x[(x)^2 - (2)^2] \\ = x(x + 2)(x - 2) \end{aligned}$$

Now

$$\begin{aligned} x^4 - 16 &= (x^2 + 4)\boxed{(x + 2)(x - 2)} \\ x^3 - 4x &= x\boxed{(x + 2)(x - 2)} \\ LCM &= [(x + 2)(x - 2)][x \cdot (x^2 + 4)] \\ LCM &= x(x + 2)(x - 2)(x^2 + 4) \\ LCM &= x[(x)^2 - (2)^2](x^2 + 4) \\ LCM &= x(x^2 - 4)(x^2 + 4) \\ LCM &= x[(x^2)^2 - (4)^2] \\ LCM &= x(x^4 - 16) \end{aligned}$$

(v) $16 - 4x^2, x^2 + x - 6, 4 - x^2$

Factorization

$$\begin{aligned} 16 - 4x^2 \\ = 4[4 - x^2] \\ = 4[(2)^2 - (x)^2] \\ = 4[(2 + x)(2 - x)] \\ = 4(2 + x)(2 - x) \end{aligned}$$

$$\begin{aligned} x^2 + x - 6 \\ = x^2 + 3x - 2x - 6 \\ = x(x + 3) - 2(x + 3) \\ = (x + 3)(x - 2) \\ = -(x + 3)(2 - x) \end{aligned}$$

$$\begin{aligned} 4 - x^2 \\ = (2)^2 - (x)^2 \\ = (2 + x)(2 - x) \end{aligned}$$

Now

$$\begin{aligned} 16 - 4x^2 &= 4\boxed{(2 + x)}\boxed{(2 - x)} \\ x^2 + x - 6 &= -(x + 3)\boxed{(2 - x)} \\ 4 - x^2 &= \boxed{(2 + x)}\boxed{(2 - x)} \\ LCM &= [(2 - x)(2 + x)][-4 \cdot (x + 3)] \\ LCM &= -4(2 - x)(2 + x)(x + 3) \\ LCM &= -4[(2)^2 - (x)^2](x + 3) \\ LCM &= -4(4 - x^2)(x + 3) \\ LCM &= 4(x^2 - 4)(x + 3) \end{aligned}$$

4. The HCF of two polynomials is $y - 7$ and their LCM is $y^3 - 10y^2 + 11y + 70$. If one of the polynomials is $y^2 - 5y - 14$, find the other.

$$\begin{aligned} HCF &= y - 7 \\ LCM &= y^3 - 10y^2 + 11y + 70 \\ p(y) &= y^2 - 5y - 14 \end{aligned}$$

Let $q(y)$ be the other polynomial

By using formula

$$\begin{aligned} p(y) \times q(y) &= HCF \times LCM \\ q(y) &= \frac{HCF \times LCM}{p(y)} \\ q(y) &= \frac{(y - 7)(y^3 - 10y^2 + 11y + 70)}{y^2 - 5y - 14} \\ &\quad y - 5 \\ y^2 - 5y - 14 &\overline{) y^3 - 10y^2 + 11y + 70} \\ &\quad \pm y^3 \mp 5y^2 \mp 14y \\ &\quad \hline &\quad -5y^2 + 15y + 70 \\ &\quad \pm 5y^2 \pm 15y \pm 70 \\ &\quad \hline &\quad 0 \\ q(y) &= (y - 7)(y - 5) \\ q(y) &= y(y - 5) - 7(y - 5) \end{aligned}$$

$$q(y) = y^2 - 5y - 7y + 35$$

$$q(y) = y^2 - 12y + 35$$

5. LCM and HCF of two polynomials $p(x)$ and $q(x)$ are $36x^3(x+a)(x^3-a^3)$ and $x^2(x-a)$ respectively. If $p(x) = 4x^2(x^2-a^2)$, find $q(x)$.

$$LCM = 36x^3(x+a)(x^3-a^3)$$

$$HCF = x^2(x-a)$$

$$p(x) = 4x^2(x^2-a^2)$$

$$q(x) = ?$$

By using formula

$$p(x) \times q(x) = LCM \times HCF$$

$$q(x) = \frac{LCM \times HCF}{p(x)}$$

$$q(x) = \frac{[36x^3(x+a)(x^3-a^3)][x^2(x-a)]}{4x^2(x^2-a^2)}$$

$$q(x) = \frac{36x^3 \cdot x^2(x+a)(x^3-a^3)(x-a)}{4x^2(x+a)(x-a)}$$

$$q(x) = 9x^3(x^3-a^3)$$

6. The HCF and LCM of two polynomials is $(x+a)$ and $12x^2(x+a)(x^2-a^2)$ respectively. Find the product of the two polynomials.

$$HCF = (x+a)$$

$$LCM = 12x^2(x+a)(x^2-a^2)$$

Let $p(x) \times q(x)$ be the product of two polynomials

By using formula

$$p(x) \times q(x) = HCF \times LCM$$

$$p(x) \times q(x) = [(x+a)][12x^2(x+a)(x^2-a^2)]$$

$$p(x) \times q(x) = [(x+a)][12x^2(x+a)(x+a)(x-a)]$$

$$p(x) \times q(x) = 12x^2(x+a)^3(x-a)$$